

Objective: I can solve equations where the inverse power results in a quadratic function, handle extraneous solutions, and manipulate literal equations.

Part 1: When the Result is Quadratic

Sometimes, eliminating the radical leaves you with an x^2 term. You must set the equation to zero and solve.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example A: Teacher Model

1. Factoring Required

$$\begin{aligned} (\sqrt[3]{x^2 - 4x})^3 &= (\sqrt[3]{12})^3 \\ x^2 - 4x &= 12 \\ x^2 - 4x - 12 &= 0 \\ (x - 6)(x + 2) &= 0 \\ \underline{x = 6 \quad x = -2} \end{aligned}$$

2. Quadratic Formula Required

$$\begin{aligned} (\sqrt[3]{x^2 - 6x})^3 &= (\sqrt[3]{9})^3 \\ x^2 - 6x &= 9 \\ x^2 - 6x - 9 &= 0 \\ a=1 \quad b=-6 \quad c=-9 \\ x &= \frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(-9)}}{2(1)} \\ &= \frac{6 \pm \sqrt{36 + 36}}{2} = \frac{6 \pm 6\sqrt{2}}{2} \\ &= \boxed{3 \pm 3\sqrt{2}} \end{aligned}$$

Practice: Resulting in Quadratics

1. Variable on Outside

$$\begin{aligned} (\sqrt{2x+15})^2 &= (x)^2 \\ 2x+15 &= x^2 \\ x^2 - 2x - 15 &= 0 \\ (x - 5)(x + 3) &= 0 \\ \underline{x = 5} \quad \cancel{x = -3} \end{aligned}$$

Since this is a
Square Root Check for
extraneous solutions

$$\begin{aligned} \sqrt{2(-3)+15} &= -3 \\ \sqrt{-6+15} &= -3 \end{aligned}$$

2. Cube Root Quadratic

$$\begin{aligned} \frac{2}{3}(x^2 + 5x)^{\frac{1}{3}} &= \frac{4}{3} \\ ((x^2 + 5x)^{\frac{1}{3}})^3 &= (2)^3 \\ x^2 + 5x &= 8 \\ x^2 + 5x - 8 &= 0 \\ a=1 \quad b=5 \quad c=-8 \\ x &= \frac{-5 \pm \sqrt{(5)^2 - 4(1)(-8)}}{2(1)} \quad 25+32 \\ &= \frac{-5 \pm \sqrt{57}}{2} \end{aligned}$$

Part 2: The Even Numerator "Two Case" Scenario

When the numerator of the exponent is **EVEN** (e.g., $\frac{2}{3}$), taking the reciprocal root creates a $\pm$ situation.

Example B

1. Standard Even Root

$$\begin{aligned} x-1 &= \pm 27 \\ x &= 1 \pm 27 \\ \boxed{x=28 \quad x=-26} \end{aligned}$$

$$\begin{aligned} (x-1)^{\frac{2}{3}} &= 9 \\ (\sqrt[3]{x-1})^2 &= \sqrt{9} \\ (\sqrt[3]{x-1})^3 &= (\pm 3)^3 \\ x-1 &= \pm 27 \end{aligned}$$

2. Complex Even Root

$$\begin{aligned} \frac{2(x^2-14)^{\frac{2}{3}}}{2} &= \frac{50}{2} \\ (x^2-14)^{\frac{2}{3}} &= 25 \\ \sqrt{(\sqrt[3]{x^2-14})^2} &= \sqrt{25} \\ (\sqrt[3]{x^2-14})^3 &= (\pm 5)^3 \\ x^2-14 &= \pm 125 \\ x^2 &= 14 \pm 125 \\ \sqrt{x^2} &= \sqrt{139} \quad \sqrt{x^2} = \sqrt{139} \\ x &= \pm \sqrt{139} \quad x = \pm \sqrt{139} \end{aligned}$$

Part 3: Literal Equations

Isolate the specific variable requested. Do not plug in numbers.

Practice: Isolating Variables

5. Biology (Kleiber's Law)

Solve this equation for m .

$$\begin{aligned} \frac{M}{K} &= k \cdot m^{\frac{3}{4}} \\ \left(\frac{M}{K}\right)^{\frac{4}{3}} &= (m^{\frac{3}{4}})^{\frac{4}{3}} \\ \left(\frac{M}{K}\right)^{\frac{4}{3}} &= m \end{aligned}$$

Part 4: Challenge Application

Planetary Motion: Kepler's Third Law states that the square of the orbital period P (in years) is equal to the cube of the average distance a (in AU) from the Sun:

$$P = a^{\frac{3}{2}}$$

Problem: A newly discovered asteroid has an orbital period P of 125 years.

- Write an equation and solve for the average distance a .

$$\begin{aligned} 125 &= (a^{\frac{3}{2}})^{\frac{2}{3}} \\ \boxed{a=25} \end{aligned}$$

- If a planet is 4 times further from the sun than the asteroid ($4a$), what is its period?

$$\begin{aligned} a &= 25 \\ 4a &= 100 \\ P &= (100)^{\frac{3}{2}} \\ \boxed{P=1000 \text{ years}} \end{aligned}$$